

MODULE : 1

INTRODUCTION TO SECURITY

Security Goals: (i) Confidentiality } def + examples
(ii) Integrity
(iii) Availability.

- He can obtain some other type information by monitoring online traffic.

- Passive attacks.

(iii) Modification of messages.

- Some portion of legitimate message is altered, or that messages are delayed or reordered, to produce an unauthorized effect.

(iv) Masquerade

- Spoofing

- One entity pretends to be a different entity.

(v) Replay:

- Attacker obtains copy of message sent by the user and later tries to replay it.

(vi) Repudiation:

- One or two parties are present: sender or receiver

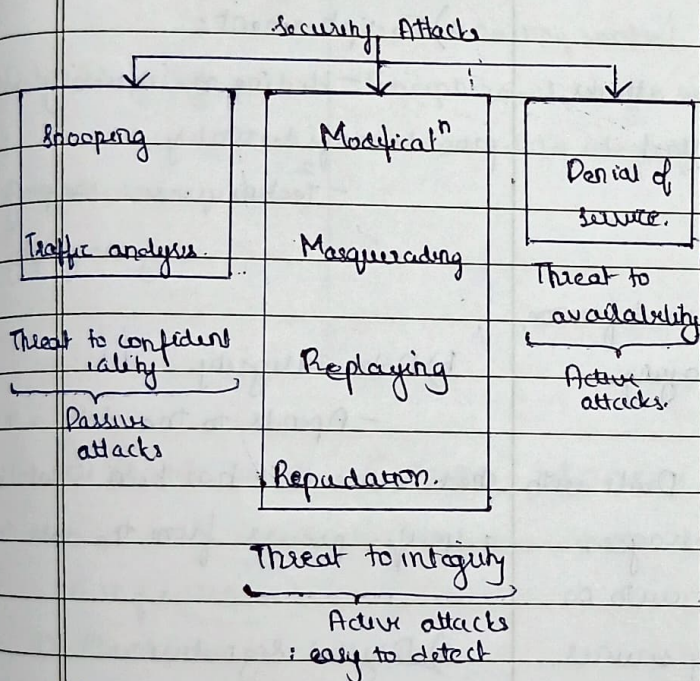
- Sender of the message might later deny that he has sent the message. The receiver of the message might later deny that he has received the message.

(vii) Denial of service:

- Prevents or inhibits the normal use or management of communication

ASPECTS OF NETWORK SECURITY

1) SECURITY ATTACKS



(i) Snooping:

- Unauthorized access to data.
- Passive attacks: affect confidentiality.

(ii) Traffic analysis.

- Masking the contents of messages or other information, traffic.

* Diff b/w passive and active attacks

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feasibility.

- It is disruption of an entire network, either by disabling the network or by overloading it with messages to degrade performance.

SECURITY SERVICES:

ITU-T has defined 5 services related to security goals and attacks.

- (i) Data confidentiality
- (ii) Authentication
- (iii) Non-repudiation
- (iv) Access control.

(i) Data Confidentiality

- Protect from passive attacks, snooping and traffic attack.

(ii) Data integrity:

- Protect data from modification, insertion, deletion and replaying.

Types

(i) Connection oriented: Deals with a stream of messages, which are sent with no

(ii) Connectionless integrity services.

(iii) Authentication.

- To verify the identity of the user / computer.

(iv) Non-repudiation:

- Protection from repudiation either the sender or the receiver of the data.

(v) Access control:

- Provides protection against

SECURITY MECHANISMS

- Security mechanisms to provide security services.

* (i) SPECIFIC SECURITY MECHANISMS

a) Encipherment:

- Hiding or covering data provide confidentiality.

Techniques: cryptography, steganography.

b) Data Integrity.

- Appends to the data a short

check value that has been created by a specific process from the data itself.

c) Digital Signature:

- The sender electronically signs the data and the receiver can electronically verify the signature.

d) Authentication exchange.

e) Traffic padding.

f) Routing control:

- Enables selection of particular physically secure routes for data and allow routing changes.

g) Notarization.

h) Access Control.

RELATIONSHIP BTW SERVICES AND MECHANISMS.

Services	Mechanisms.
Data Confidentiality:	Encipherment and routing control
Data integrity	Encipherment, digest sig nature, data integrity.
Authentication	

(iii) Secret key:

- Secret key is also i/p to encryption algorithm. The key is value independent of plaintext and algorithm.

(iv) Ciphertext:

- Scrambled message produced as o/p. Depends on plaintext and secret key.

(v) Decryption algorithm.

- Encryption algorithm run in

Cryptanalysts:

- Attack rely on nature of algorithm, also on plaintext.
- Exploits characteristics of algorithm.

Brute force attacks:

- Attacks rely on keys.

SYMMETRIC CIPHER MODEL:

- A symmetric encryption scheme has 5 ingredients.

(i) Plaintext:

- Original intelligible p message or data that is fed into the algorithm as input.

(ii) Encryption algorithm:

- It performs various substitutions and transformations on the plaintext.

HILL CIPHER:

- It is a polygraphic substitution cipher based on linear algebra.

Encryption

$$C = KP \text{ mod } 26$$

Decryption

$$P = K^{-1}C \text{ mod } 26$$

Q: Encrypt the plain text "attack" using Hill cipher for the given key

0	1	2	3	4	5	6	7	8	9	10	11	12	13
a	b	c	d	e	f	g	h	i	j	k	l	m	n
14	15	16	17	18	19	20	21	22	23	24	25		
o	p	q	r	s	t	u	v	w	x	y	z		

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$$\begin{bmatrix} 2 & 3 \\ 3 & 6 \end{bmatrix} \quad k = \begin{bmatrix} 2 & 3 \\ 3 & 6 \end{bmatrix}$$

Pls "attack"

$$C = kP \text{ mod } 26$$

It is a 2x2 matrix, convert the vector to length 2.

$$a|k|a|ck$$

$$\Rightarrow 1^{\text{st}} \text{ vector: } \begin{bmatrix} a \\ t \end{bmatrix} = \begin{bmatrix} 0 \\ 19 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 2 & 3 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} 0 \\ 19 \end{bmatrix} \text{ mod } 26$$

$$\Rightarrow \begin{bmatrix} 57 \\ 114 \end{bmatrix} \text{ mod } 26$$

$$\Rightarrow \begin{bmatrix} 5 \\ 10 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} f \\ k \end{bmatrix} = \begin{bmatrix} f \\ k \end{bmatrix}$$

$$\Rightarrow 2^{\text{nd}} \text{ vector} \Rightarrow \begin{bmatrix} t \\ a \end{bmatrix} = \begin{bmatrix} 19 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 2 & 3 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} 19 \\ 0 \end{bmatrix} \text{ mod } 26$$

$$\Rightarrow \begin{bmatrix} 38 \\ 57 \end{bmatrix} \text{ mod } 26$$

$$\Rightarrow \begin{bmatrix} 12 \\ 5 \end{bmatrix} = \begin{bmatrix} m \\ f \end{bmatrix}$$

$$\Rightarrow 3^{\text{rd}} \text{ vector} \Rightarrow \begin{bmatrix} c \\ k \end{bmatrix} = \begin{bmatrix} 2 \\ 10 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 2 & 3 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} 2 \\ 10 \end{bmatrix}$$

$$= \begin{bmatrix} 34 \\ 66 \end{bmatrix} \text{ mod } 26$$

$$\Rightarrow \begin{bmatrix} 8 \\ 14 \end{bmatrix} = \begin{bmatrix} I \\ O \end{bmatrix}$$

$$\Rightarrow \text{"attack"} = \text{"FKMFIO"} = C$$

Decryption using same key

$$P = k^{-1}C \text{ mod } 26$$

$$k = \begin{bmatrix} 2 & 3 \\ 3 & 6 \end{bmatrix}$$

$$k^{-1} = \begin{bmatrix} 6 & -3 \\ -3 & 2 \end{bmatrix} / 3$$

$$(i) P = \begin{bmatrix} 6 & -3 \\ -3 & 2 \end{bmatrix} / 3$$

We need to remove -ve value inside the matrix.

$$\Rightarrow -3 \Rightarrow 26 - 3 = 23$$

$$\Rightarrow k^{-1} = \begin{bmatrix} 6 & 23 \\ 23 & 2 \end{bmatrix} / 3$$

As $2/3$ gives a decimal and there is no alphabet with decimal so we need to remove such mat

$$3 \times 1 = 1$$

But $3 \times \text{mod } 26 \rightarrow$ is present. so $1/3$ is not the value.

so $2 \times \text{mod } 26 = 1$ should be obtained for that

$$3 \times 9 \text{ mod } 26 = 27 \text{ mod } 26$$

gives 1.

$$k^{-1} \Rightarrow 9 \times \begin{bmatrix} 6 & 23 \\ 23 & 2 \end{bmatrix} \text{ mod } 26$$

$$K^{-1} \Rightarrow \begin{bmatrix} 54 & 207 \\ 207 & 18 \end{bmatrix} \text{mod } 26$$

$$t^{-1} = \begin{bmatrix} 2 & 25 \\ 25 & 18 \end{bmatrix}$$

$$\Rightarrow P_1 = \begin{bmatrix} 2 & 25 \\ 25 & 18 \end{bmatrix} \begin{bmatrix} 5 \\ 10 \end{bmatrix} \text{mod } 26$$

$$= \begin{bmatrix} 260 \\ 3085 \end{bmatrix} \text{mod } 26$$

$$= \begin{bmatrix} 0 \\ 19 \end{bmatrix} = \begin{bmatrix} a \\ t \end{bmatrix} \quad \underline{at}$$

$$P_2 = \begin{bmatrix} 2 & 25 \\ 25 & 18 \end{bmatrix} \begin{bmatrix} 12 \\ 5 \end{bmatrix} \text{mod } 26$$

$$= \begin{bmatrix} 149 \\ 390 \\ 200 \end{bmatrix} \text{mod } 26 = \begin{bmatrix} 19 \\ 10 \end{bmatrix} = \begin{bmatrix} t \\ a \end{bmatrix}$$

ta

$$P_3 = \begin{bmatrix} 2 & 25 \\ 25 & 18 \end{bmatrix} \begin{bmatrix} 9 \\ 14 \end{bmatrix} \text{mod } 26$$

$$= \begin{bmatrix} 366 \\ 452 \end{bmatrix} \text{mod } 26 = \begin{bmatrix} 2 \\ 10 \end{bmatrix} = \begin{bmatrix} a \\ t \end{bmatrix}$$

at

Decrypted code = "attack"

Q: Encrypt the plaintext "safe message" using Hill cipher for the given key "ciphering"

Ans: $P = \text{"safe message"}$

$K = \text{"ciphering"}$

As K has 9 letters, so make a 3×3 matrix.

$$\Rightarrow 3 \times 3 \text{ matrix} = \begin{bmatrix} c & i & p \\ h & e & a \\ i & n & g \end{bmatrix} =$$

$$K = \begin{bmatrix} 2 & 8 & 15 \\ 7 & 4 & 17 \\ 8 & 13 & 6 \end{bmatrix}$$

$\Rightarrow$ grouping the messages into 3 letters.

$$\Rightarrow 1^{\text{st}} \text{ vector} \Rightarrow \begin{bmatrix} s \\ a \\ t \end{bmatrix} = \begin{bmatrix} 18 \\ 0 \\ 5 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 2 & 8 & 15 \\ 7 & 4 & 17 \\ 8 & 13 & 6 \end{bmatrix} \begin{bmatrix} 18 \\ 0 \\ 5 \end{bmatrix} \text{mod } 26$$

$$= \begin{bmatrix} 111 \\ 211 \\ 174 \end{bmatrix} \text{mod } 26$$

$$= \begin{bmatrix} 7 \\ 9 \\ 18 \end{bmatrix} = \begin{bmatrix} h \\ o \\ s \end{bmatrix} = \text{HDS}$$

$$\Rightarrow 2^{\text{nd}} \text{ vector} \Rightarrow \begin{bmatrix} E \\ M \\ E \end{bmatrix} = \begin{bmatrix} 4 \\ 12 \\ 4 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 2 & 8 & 15 \\ 7 & 4 & 17 \\ 8 & 13 & 6 \end{bmatrix} \begin{bmatrix} 4 \\ 12 \\ 4 \end{bmatrix} \text{mod } 26$$

$$= \begin{bmatrix} 164 \\ 144 \\ 212 \end{bmatrix} \text{mod } 26 = \begin{bmatrix} 18 \\ 14 \\ 4 \end{bmatrix} = \text{IOE}$$

$$\Rightarrow 3^{\text{rd}} \text{ vector} \Rightarrow \begin{bmatrix} s \\ s \\ n \end{bmatrix} = \begin{bmatrix} 18 \\ 18 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 2 & 8 & 15 \\ 7 & 4 & 17 \\ 8 & 13 & 6 \end{bmatrix} \begin{bmatrix} 18 \\ 18 \\ 0 \end{bmatrix} \text{mod } 26 = \begin{bmatrix} 180 \\ 198 \\ 378 \end{bmatrix} \text{mod } 26$$

$$= \begin{bmatrix} 24 \\ 16 \\ 14 \end{bmatrix} = \begin{bmatrix} Y \\ Q \\ O \end{bmatrix} = \underline{\underline{YQO}}$$

Hth vector = $\begin{bmatrix} G \\ E \\ S \end{bmatrix}$

(Incomplete).

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Decryption.

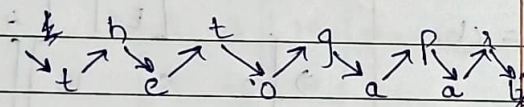
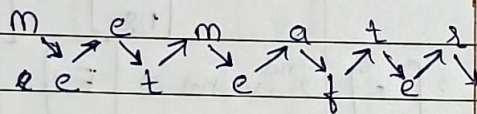
first string { W E C R U O
second string { ER DS OE ER NT
A I V D A

ONE TIME PAD:

TRANSPOSITION TECHNIQUES:

Q: Cipher the message: "meet me
after the toga party" with ^{thru} depth 2

Ans: meet me at the toga party



Ciphertext:

① t n a ② a p t m ③ t s u o ④ a d d w ⑤ c o i x
⑥ t n l y ⑦ p e t z

Q: Decrypt the following message
that was encrypted using a rail fence
cipher with 4 rails.

mematahtgpye tefeteoat

TTPT QDSTP RSKII XEDOH
EIUNS ESLDY TEMES SERSE

Q: Encrypt the message: "WE ARE DISCOV-
ERED, RUN AT ONCE" with 3 rails

NEL SC NEAUC FLERE GAMAE
BHDIH SCUCD NG.

